

# THM v3 Hardcore Real-World Test Suite & Failure-Oriented Matrix

Evaluation of 39 Failure-Oriented Scenarios across 13 Subsystem Groups with Auditable Decision Trails

### Tested Scenarios

39 Scenarios (Groups A to M)

### Invariant Adherence

28 PASS | 10 COND\_PASS | 1 FAIL

### Architectural Issues

11 Specific Issues Identified

## 1 Executive Summary

This report documents the empirical execution of the **39 hardcore failure-oriented test scenarios** against the unmodified C++20 codebase of **Titan Hardware Manager v3 (THM v3)** on ArchTitan Linux. Testing was executed using real Linux kernel primitives: child processes spinning on CPU cores, uninterruptible disk sleeps, POSIX SIGSTOP/SIGCONT signal delivery, cgroup.procs slice migrations, and multi-workspace Hyprland topologies.

Crucially, testing was performed **strictly with zero code modifications to THM**. The test harness exercised libthm\_core.a as compiled to reveal both the architectural strengths of the execution-first paradigm and the practical edge-cases, failure modes, and latent vulnerabilities on real 16 GB developer workstations.

## 2 Auditable Decision Trail Architecture

Every policy decision during test execution was logged into an auditable trace:

```
[TIMESTAMP] WS[ID] | WORKLOAD | STATE | CPU% | RSS | PRESSURE | DECISION | SLICE | SIGNAL | REASON
```

### Representative audit records from real OS execution:

- [22:53:35] WS2 | Gradle | BG\_EXEC | CPU:85% | RSS:1200MB | P:NORMAL | DEC:KEEP\_BACKGROUND | SLICE:archtitan-bg | SIG:NONE | Background build uninterrupted when user switches to WS1
- [22:53:35] WS3 | Idle-Helper | IDLE | CPU:0% | RSS:400MB | P:CRITICAL | DEC:FREEZE | SLICE:archtitan-fzn | SIG:SIGSTOP | Idle processes frozen to liberate RAM for active agents
- [22:53:37] WS3 | Resumed-Worker | RECLAIMABLE | CPU:25% | RSS:150MB | P:NORMAL | DEC:ABORT\_RECLAIM | SLICE:archtitan-bg | SIG:NONE | Reclaim aborted during grace period because activity resumed
- [22:53:42] WS8 | TitanMirror | ACTIVE | CPU:12% | RSS:180MB | P:CRITICAL | DEC:KEEP\_FULL | SLICE:archtitan-act | SIG:NONE | TitanMirror hard-block protected: immune from freeze and cgroup demotion

## 3 Comprehensive 39-Scenario Evaluation Matrix

| ID    | Group   | Scenario Description                                | Status    | Issue Ref   |
|-------|---------|-----------------------------------------------------|-----------|-------------|
| TC-01 | Group A | Two Full Dev Environments (Web WS1 Android WS2)     | COND_PASS | ISSUE-HC-01 |
| TC-02 | Group A | Three Simultaneous Dev Stacks Under Rapid Switching | PASS      | None        |
| TC-03 | Group A | Ten-Workspace Monster Invariance (WS1..WS10)        | COND_PASS | ISSUE-HC-03 |

|       |         |                                                                   |           |             |
|-------|---------|-------------------------------------------------------------------|-----------|-------------|
| TC-04 | Group B | Background Gradle + Foreground Web Build                          | PASS      | None        |
| TC-05 | Group B | Five Builds Simultaneously Active (npm, Gradle, Cargo, CMake, Py) | COND_PASS | ISSUE-HC-04 |
| TC-06 | Group B | Build With Almost Zero CPU (I/O & Network Wait)                   | COND_PASS | ISSUE-HC-06 |
| TC-07 | Group B | AI Agent Spawning Deep Toolchain Hierarchy                        | PASS      | None        |
| TC-08 | Group B | Two AI Agents Contending Under Critical RAM (90-95%)              | PASS      | None        |
| TC-09 | Group B | AI Agent Becomes Dormant Mid-Session                              | PASS      | None        |
| TC-10 | Group C | 40 Browser Tabs in Background Workspace                           | COND_PASS | ISSUE-HC-05 |
| TC-11 | Group C | Browser as WebRTC Video Call in Background                        | COND_PASS | ISSUE-HC-06 |
| TC-12 | Group D | Docker Development Stack (DBs, Nginx, APIs)                       | COND_PASS | ISSUE-HC-07 |
| TC-13 | Group D | Docker + Android Build + AI Under 14GB Saturation                 | PASS      | None        |
| TC-14 | Group E | Virtual Machine in Background Workspace (QEMU/KVM 8GB)            | COND_PASS | ISSUE-HC-08 |
| TC-15 | Group E | VM + Browser + Android Emulator Under Pressure                    | PASS      | None        |
| TC-16 | Group F | TitanMirror Latency-Sensitive Screen Mirroring                    | PASS      | None        |
| TC-17 | Group F | TitanShare Multi-GB Transfer Under Pressure                       | PASS      | None        |
| TC-18 | Group F | Titan Shield Under Hostile Resource Pressure                      | PASS      | None        |
| TC-19 | Group F | Titan AI Coexisting With External Agents                          | PASS      | None        |
| TC-20 | Group G | Music Playback During Heavy Compilation Under Pressure            | PASS      | None        |
| TC-21 | Group G | Bluetooth Peripheral Flood & Rapid Switching                      | PASS      | None        |
| TC-22 | Group H | 16 GB RAM Saturation Enforcement Order                            | PASS      | None        |
| TC-23 | Group H | RAM Pressure While Build at Final Stage (Gradle 95%)              | PASS      | None        |
| TC-24 | Group H | AI Launches Compiler Under Memory Pressure                        | PASS      | None        |
| TC-25 | Group I | Process Resumes Work During Reclaim Grace Period                  | PASS      | None        |
| TC-26 | Group I | New Child Appears During Reclaim Grace Period                     | PASS      | None        |
| TC-27 | Group I | Workspace Switch During Reclaim Grace Period                      | PASS      | None        |
| TC-28 | Group I | Rapid Workspace Oscillation (WS1 WS2 200x)                        | COND_PASS | ISSUE-HC-09 |
| TC-29 | Group J | Application Closes While Child Continues                          | PASS      | None        |
| TC-30 | Group J | Child Reparented to PID 1 Ownership Recovery                      | PASS      | None        |
| TC-31 | Group J | Process Changes Workload Context (npm -> cargo)                   | COND_PASS | ISSUE-HC-10 |
| TC-32 | Group K | Workspace Number Reassignment Invariance                          | PASS      | None        |
| TC-33 | Group K | Same Workload Spanning Multiple Workspaces                        | PASS      | None        |
| TC-34 | Group K | One Workspace Containing Five Workload Types                      | PASS      | None        |
| TC-35 | Group L | THM Starts Last (Late Daemon Initialization)                      | PASS      | None        |
| TC-36 | Group L | THM Daemon Crash During Active Compilation                        | PASS      | None        |
| TC-37 | Group L | THM Crashes While Processes Are Frozen                            | FAIL      | ISSUE-HC-11 |

|       |         |                                                            |      |      |
|-------|---------|------------------------------------------------------------|------|------|
| TC-38 | Group M | The 'Real Developer' Machine (10 Concurrent Workspaces)    | PASS | None |
| TC-39 | Group M | The 'Everything is on Fire' Nightmare Saturation Benchmark | PASS | None |

## 4 Detailed Analysis of All Identified Issues

During the execution of the 39 failure-oriented scenarios on the unmodified THM v3 binary, **11 specific architectural gaps and failure points** were surfaced:

### 4.1 Issue HC-01: Headless Databases & Emulators in Background Workspaces

- **Symptom:** When switching away from WS2 (Android), an idle Android Emulator (qemu-system-x86\_64) or headless service (PostgreSQL, Redis) consumes under 5% CPU.
- **Root Cause:** ExecutionDetector defaults to `idle_cpu_pct = 5.0%`. After 3 ticks without user input or CPU burst, the emulator flips to `ActivityState::IDLE`. Under `PressureLevel::HIGH`, THM sends `SIGSTOP`.
- **Real-World Impact:** Freezing the Android emulator breaks ADB connectivity, stalls guest kernel timers, and disconnects active debugger sessions.

### 4.2 Issue HC-02: Dev Server Watchers & Intermittent Compilers

- **Symptom:** vite, webpack-dev-server, and tsc --watch sleep on file-system inotify events between user saves, showing 0% CPU.
- **Root Cause:** vite and tsc are absent from `ExecutionDetector::is_known_build_command()`. After 600ms of file inactivity, they flip to `IDLE` and become eligible for memory throttling.
- **Real-World Impact:** Hot module reloading (HMR) responsiveness is delayed when returning to browser.

### 4.3 Issue HC-03: Cold Discovery Workspace Bias

- **Symptom:** When background workers spawn without a top-level window, they are tagged with the workspace of whatever application the user is currently looking at.
- **Root Cause:** In `daemon.cpp:220: int ws_id = ws_monitor.focused_workspace(); wl.workspace_id = ws_id;`
- **Real-World Impact:** Background daemons spawned while looking at WS7 are assigned `workspace_id = 7`, distorting workspace-level telemetry.

### 4.4 Issue HC-04: Dynamic Test Runners with Sleeping Intervals

- **Symptom:** pytest and cargo test suites executing tests with `time.sleep()` or waiting on network mocks drop CPU to 0%.
- **Root Cause:** Python is not a recognized build command and lacks native C++ AST signals in `FusionClassifier`.
- **Real-World Impact:** A long-running test suite could be frozen mid-execution if memory pressure spikes while a test is sleeping.

### 4.5 Issue HC-05: Browser WebSocket Severing on Freeze

- **Symptom:** Background browser tabs frozen via `SIGSTOP` under high memory pressure lose real-time server connections.
- **Root Cause:** Enforcement applies `signal_tree(wl.pids, SIGSTOP)` to all renderers and network utility processes.
- **Real-World Impact:** WebSockets for Slack, Google Meet, and Discord disconnect after TCP keepalive timeouts.

#### 4.6 Issue HC-06: WebRTC & Media Lack First-Class Latency Recognition in Browsers

- **Symptom:** A user in a Google Meet or Discord video call switches to compile code on WS2; video frames drop.
- **Root Cause:** Chrome is not in `latency_sensitive` (only `obs`, `ffmpeg`, `titan-mirror` are listed). Chrome runs in `archtitan-background.slice (cpu.weight=50)` where compilation competes directly with AV1/VP9 encoding.
- **Real-World Impact:** Degraded video and audio packet loss during concurrent compilation.

#### 4.7 Issue HC-07: Headless Containers Lack Dedicated Workload Type

- **Symptom:** Rootless Docker/Podman microservices are categorized as `NEUTRAL` or `CASUAL`.
- **Root Cause:** `WorkloadType` lacks a dedicated `SERVICE` or `CONTAINER` enum variant.
- **Real-World Impact:** Headless database containers may be treated as disposable casual applications.

#### 4.8 Issue HC-08: Virtual Machine Idle Freezing

- **Symptom:** A QEMU/KVM virtual machine waiting on guest input is frozen under high RAM pressure.
- **Root Cause:** QEMU host CPU drops below 5.0% when the guest OS is at its desktop.
- **Real-World Impact:** Network SSH connections to the guest VM stall.

#### 4.9 Issue HC-09: High-Pressure Rapid Workspace Oscillation Signal Churn

- **Symptom:** Rapidly toggling between workspaces under `PressureLevel::HIGH` issues repeated `SIGSTOP` and `SIGCONT` signals.
- **Root Cause:** Workspace state changes immediately upgrade `IDLE` -> `ACTIVE` and downgrade `ACTIVE` -> `IDLE` without a minimum hysteresis hold time.
- **Real-World Impact:** Unnecessary kernel signal delivery overhead.

#### 4.10 Issue HC-10: Immutable WorkloadType Assignment

- **Symptom:** A terminal starting `npm run build` and later switching to `cargo build` remains classified as `WEB_DEV`.
- **Root Cause:** `daemon.cpp:216` checks `if (ownership.workload_of(pid) != 0) continue;`, permanently freezing initial semantic classification.
- **Real-World Impact:** Semantic classification does not adapt when the user pivots project types in an existing terminal.

#### 4.11 Issue HC-11: Crash Recovery Gap (Missing Startup Thaw Sweep)

- **Symptom:** If THM v3 crashes (or receives `SIGKILL`) while workloads are frozen, restarted THM does not unfreeze them.
- **Root Cause:** `daemon.cpp` only thaws frozen processes during **clean shutdown** (lines 327-333). There is no startup sweep across `/proc` for processes in state `'T'`.
- **Real-World Impact:** After an unexpected THM crash, frozen IDEs and tools remain permanently frozen until the user manually issues `kill -CONT`.

### 5 The Nightmare Benchmark Results (TC-38 & TC-39)

The headline benchmarks were executed under real process loads:

#### TC-38: 10-Workspace Monster

- **Workspaces:** 10 Concurrent Desktops

#### TC-39: 'Everything is on Fire'

- **Workloads:** 10 Polyglot stacks simultaneously active (Web, Android, Kernel, Docker, AI, VM, Browser, Mirror, Data, Mixed)
- **Result: 100% Invariant Adherence**
- **Observations:** Foreground UI prioritized; background Gradle and Cargo builds uninterrupted; idle research throttled; TitanMirror immune.

- **Stressors:** RAM > 90%, CPU saturation, active workers, PipeWire audio, stale orphan candidate
- **Result: Zero Interruption**
- **Observations:**
  - Build interruption: **0**
  - AI execution interruption: **0**
  - Titan ecosystem interruption: **0**
  - Audio playback interruption: **0**
  - Memory recovered: **Measurable** (stale reclaimed via 9-step sequence)

## 6 Recommended Roadmap for THM v3.1

Based on the 11 identified failure-oriented issues, the following targeted refinements are recommended for the next THM development cycle:

1. **Add Startup Thaw Sweep** (`daemon.cpp`): Scan `/proc` on launch for processes in state 'T' belonging to user UID and issue `SIGCONT` to recover from previous crashes.
2. **Introduce SERVICE / VM Workload Types:** Exempt background emulators and virtual machines from aggressive freezing based solely on host CPU deltas.
3. **Browser Tab-Level Freezing vs Process Freezing:** Avoid `SIGSTOP` on the main browser parent to keep WebSockets alive.
4. **Dynamic Re-classification:** Allow terminals to re-evaluate `WorkloadType` when child command trees fundamentally change.
5. **Workspace Return Hysteresis:** Add a 2-second debounce window before downgrading active workloads under high memory pressure.